

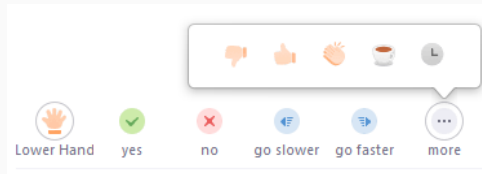
Algorithms for Lattice Fermions User Workshop 2022

14-18th Feb 2022

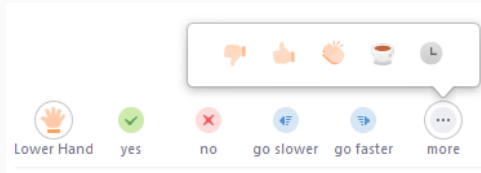
Welcome



1. Please **raise** your hands:



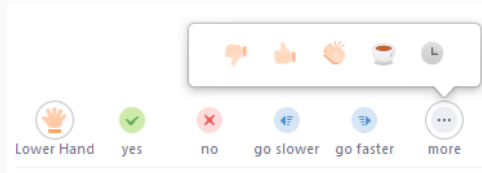
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2. Keep yourself **muted** during presentations in order to avoid background noise.

Organizational remarks

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3. Got any questions/comments? Please use the Zoom **chat**.
4. If you encounter any issues, please let us know in the chat, in Discord (<https://discord.gg/VppWWEPMHa>), or send an email to alf@physik.uni-wuerzburg.de

1. Organizational remarks

2. The ALF Project

ALF Features

ALF Team

ALF Timeline & Sponsors

3. Workshop Program

Algorithms for Lattice Fermions: general code for auxiliary-field quantum Monte Carlo simulations

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General Hamiltonian

- single-body operators,
- squares of single-body operators
- single-body operators coupled to a scalar field

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- observables (equal-time and time-displaced)

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- near-optimal single-core performance
- restart facilities
- DHF5 data files

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- extensive documentation
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Advanced Features

- parallel tempering
- ground state projective QMC
- global Monte Carlo updates
- continuous fields
- Langevin dynamics updates
- stochastic Maximum Entropy
- symmetric Trotter decomposition
- Dimer-Dimer correlations
- Cotunneling (Kondo)
- Rényi Entropy

ALF Collaboration

ALF Team members:



Fakher Assaad



Martin Bercx



Florian Goth



Anika Götz



Johannes Hofmann



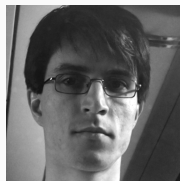
Emilie Huffman



Zihong Liu



Francesco Parisen
Toldin



Jefferson S.E.
Portela



Jonas Schwab

A short history of ALF:

2016 ALF 0.5

2017 ALF 1.0

M. Bercx, F. Goth, J. S. Hofmann, and F. F. Assaad, SciPost Phys. **3** (2017), 013.

2020 ALF 2.0, pyALF

preprint: <https://arxiv.org/abs/2012.11914> (accepted in SciPost Phys. Codebases)

2021 ALF 2.1, ALF 2.2, pyALF 2.1

2022 pyALF 2.2



SFB1170
ToCoTronics



Leibniz-Rechenzentrum
der Bayerischen Akademie der Wissenschaften



GCS
Gauss Centre for Supercomputing



Center of excellence – complexity and
topology in quantum matter



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A joint talk of the Saha Institute and ALF Workshop: <https://meet.google.com/itk-qigc-vei>
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 - Participants' presentations: ??

Workshop Program

	Friday 11/02	Monday 14/02	Tuesday 15/02	Wednesday 16/02	Thursday 17/02	Friday 18/02
(GMT+1)						
10:00	ALF / pyALF Installation				Workgroup	Workgroup
11:00			Tutorial	Tutorial	Contributed Talk	Contributed Talk
12:00						
-			Tutorial	Tutorial	Workgroup	Workgroup
13:00			ALF basics: <i>Observ.</i>	ALF basics: <i>Hopp.</i>	Advanced pyALF	
-						
14:00		Welcome	Break Room	Break Room	Break Room	Workgroup Presentations
-		Kick-off Talk	Fermion Monte Carlo + Project Proposals	Contributed Talks	Contributed Talks	
15:00		Fermion Monte Carlo				
-			Break Room	Break Room	Break Room	
16:00		Break Room	Tutorial: <i>Advanced</i>	Fermion Monte Carlo + Project Proposals (2)	Discussion	
-		Tutorial: <i>Intro.</i>				
17:00		ALF basics: <i>Overview & Lattices</i>	ALF basics: <i>Interact.</i>	ALF basics: <i>Models & Your own model</i>	Workgroup	Discussion
-						Break Room
18:00	ALF / pyALF Installation	Tutorial	Tutorial	Workgroup		
(GMT+1)						

Pre-Conference

Getting to Know ALF

ALF in Production

Key:

Plenary/synchronous

Break-out rooms/on-call

ALF *basics*